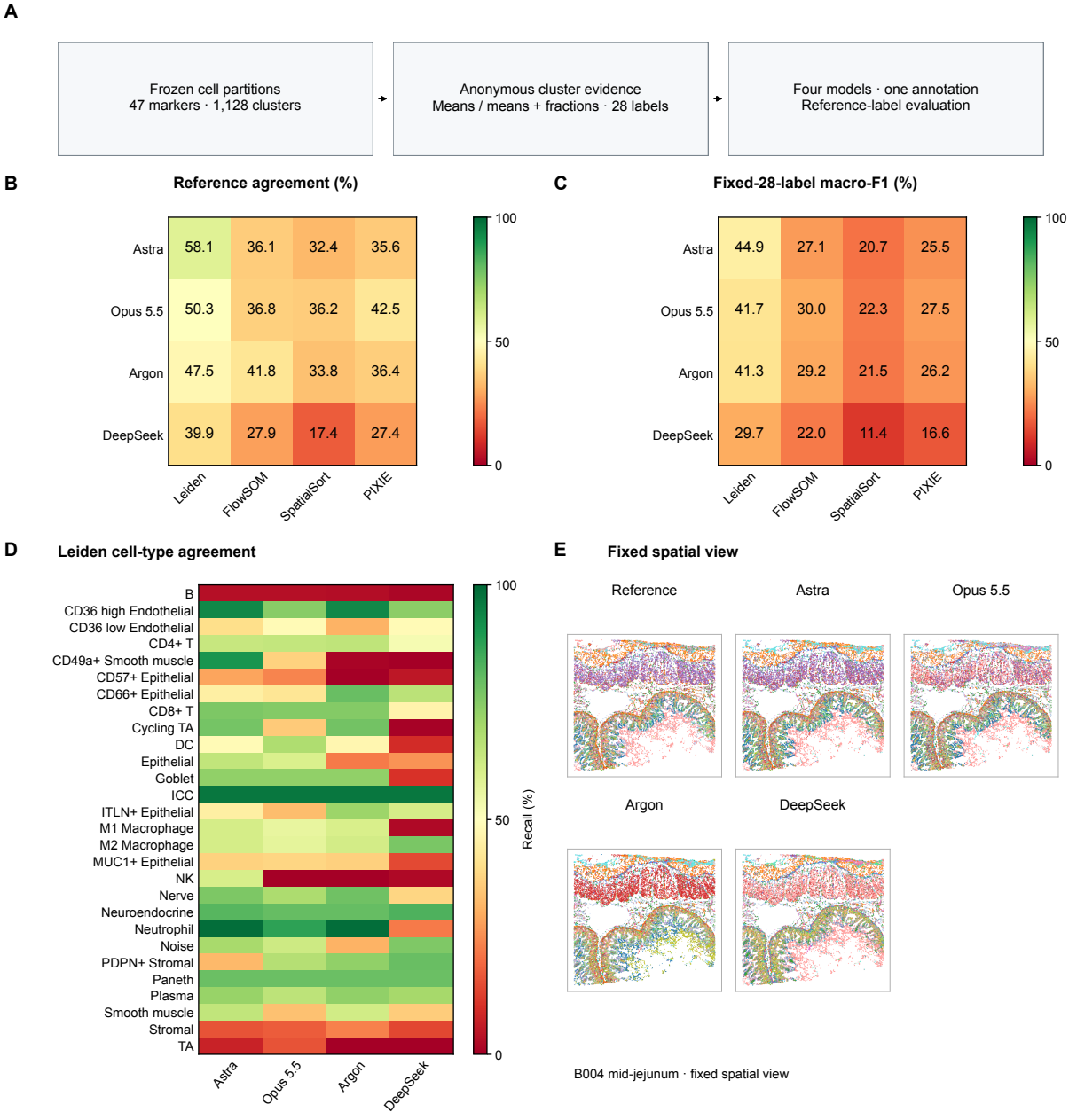

A
Frozen cell partitions
47 markers · 1,128 clusters
Anonymous cluster evidence
Means / means + fractions · 28 labels
Four models · one annotation
Reference-label evaluation
B
Reference agreement (%)
Astra
Opus 5.5
Argon
DeepSeek
58.1
36.1
32.4
35.6
50.3
36.8
36.2
42.5
47.5
41.8
33.8
36.4
39.9
27.9
17.4
27.4
Leiden
FlowSOM
SpatialSort
PIXIE
100
50
0
C
Fixed-28-label macro-F1 (%)
44.9
27.1
20.7
25.5
41.7
30.0
22.3
27.5
41.3
29.2
21.5
26.2
29.7
22.0
11.4
16.6
D
Leiden cell-type agreement
B
CD36 high Endothelial
CD36 low Endothelial
CD4+ T
CD49a+ Smooth muscle
CD57+ Epithelial
CD66+ Epithelial
CD8+ T
Cycling TA
DC
Epithelial
Goblet
ICC
ITLN+ Epithelial
M1 Macrophage
M2 Macrophage
MUC1+ Epithelial
NK
Nerve
Neuroendocrine
Neutrophil
Noise
PDPN+ Stromal
Paneth
Plasma
Smooth muscle
Stromal
TA
Recall (%)
E
Fixed spatial view
Reference
Astra
Opus 5.5
Argon
DeepSeek
B004 mid-jejunum · fixed spatial view